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Abstract

The kinematics of the Taurus molecular cloud complex have so far mainly been studied through the discrete velocities of its individual members. A full three-dimensional space velocity, however, requires a radial velocity, which is available for only about one third of the known members, while proper motions are measured for essentially all of them. This thesis therefore reconstructs a continuous three-dimensional velocity field of the complex from the full stellar sample, using information field theory as implemented in `NIFTy`. Each velocity component is modeled as a correlated field on a grid of 3 pc cells spanning $126 \times 69 \times 81$ pc, with a power spectrum inferred from the data. The input catalog consists of 557 members identified with `SigMA` in Gaia DR3 astrometry, of which 123 contribute a radial velocity after quality cuts. The reconstruction reproduces the bulk motion of the complex and resolves velocity differences between its subgroups and recovers an age–velocity gradient that is already visible in the stellar data. Both the mean motion and the quantities derived from the field, however, depend strongly on the individual run of the algorithm: repetitions with a different random seed, with different quality cuts or with a modified model yield mean motions that scatter by several $\mathrm{km\,s^{-1}}$, while the posterior uncertainty within a single run stays far below that. A comparison of several such runs is therefore used to separate features demanded by the data from artifacts of a particular reconstruction. The residuals between the stars and the field expose an important limitation of the model: the peculiar motion of a star, which is unrelated to the underlying flow, has nowhere to go but into the field itself, so the motion of single stars is imprinted on a field that is meant to describe the motion of the complex. Granting the stars a common intrinsic velocity dispersion, inferred alongside the field, improves quality statistics like residuals, but does so by widening the uncertainties rather than by bringing the field closer to the data.

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1. Introduction

Our Galaxy, the Milky Way, is a barred spiral galaxy made up of a central bulge, a dominant stellar disk and a diffuse stellar halo (Figure 1.1). The Sun sits inside the disk at a distance of approximately 8 kpc from the Galactic center, about halfway out to its edge at a radius of roughly 15 kpc [8]. Everything discussed in this thesis happens within a few hundred parsecs of the Sun, a fraction of its distance to the Galactic center.

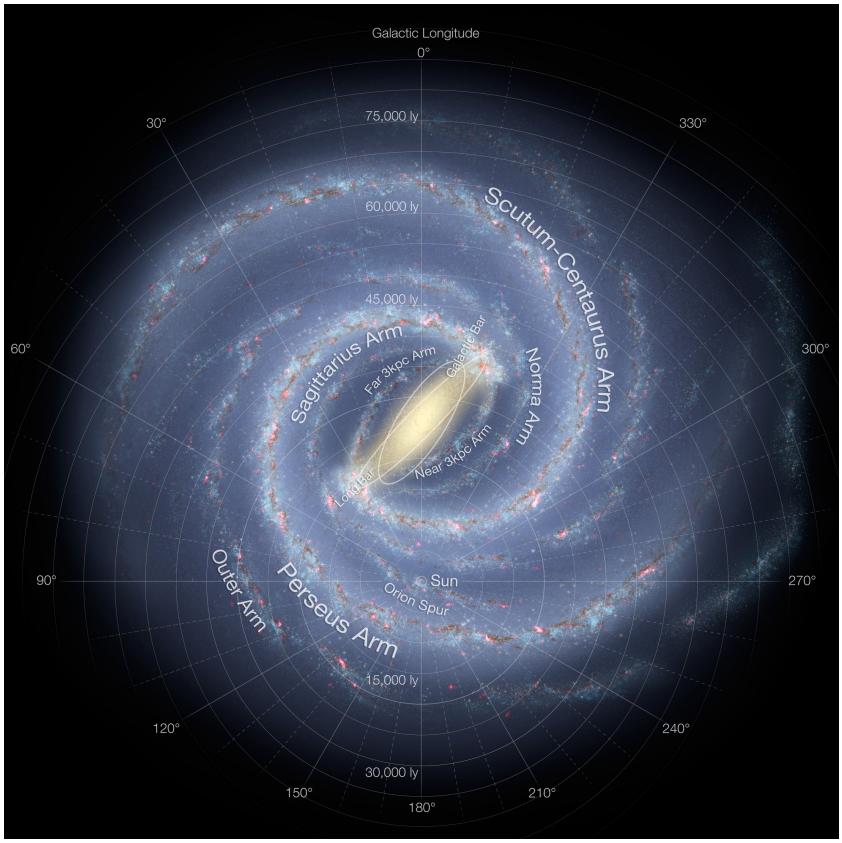


Figure 1.1.: Artist's impression of the Milky Way seen face-on, with the spiral arms, the central bar and the position of the Sun indicated. Rings mark the distance from the Sun in light years, radial lines the Galactic longitude. The region studied in this thesis lies within a few hundred parsecs of the Sun and is far smaller than any structure labelled here. Image credit: NASA/JPL-Caltech/ESO/R. Hurt [28].

Within this patch, the Sun lies close to the center of a large cavity of hot, low-density gas that is bounded by a shell of cold gas and dust with a present-day radius of about 165 pc: the Local Bubble [37]. Almost all young star-forming regions near the Sun are found on the surface of this shell rather than inside the cavity, a connection that makes the local neighborhood a valuable laboratory for studying supernova-driven star formation.

One of these regions on the surface of the Local Bubble is the Taurus molecular cloud (TMC), which is the subject of this thesis. Taurus is one of the most intensively studied star-forming regions in the Milky Way. Its close proximity to the Sun allows young stellar objects (YSOs) to be spatially resolved and characterized individually across the full complex. Unlike denser, high-mass regions such as the Orion Nebula Cluster, Taurus is an example of low-density star formation [25], making it possible to study the formation of solar-type stars in relative isolation. Its stellar population spans the full range of early evolutionary stages, from deeply embedded protostars to more evolved, disk-free pre-main-sequence stars [26].

The aim of this thesis is to study the kinematics of the Taurus star-forming region. So far, this has mainly been done by looking at the discrete, scattered velocities of individual stars. However, a full 3D space velocity can only be obtained for stars that also have a measured radial velocity, which is not the case for every member of the complex (only about one third), so for many stars only the two-dimensional proper motion is known. Discarding those stars would mean setting aside two thirds of the sample, so this thesis instead seeks a clearer picture by inferring a three-dimensional velocity field from all the available stellar velocity data. This velocity field is reconstructed using Numerical Information Field Theory (NIFTy), a Bayesian framework for field inference that models the continuous velocity field on a three-dimensional grid. The input data consist of three-dimensional positions, proper motions and, where available, radial velocities all from the latest Gaia DR3 measurements [18]. The resulting velocity field allows further analysis of both the motion of the whole Taurus complex and the internal kinematics of its substructures.

The thesis is structured as follows: Chapter 2 gives a short introduction to already present knowledge about the TMC and its kinematics. Chapter 3 briefly explains the methods used to reconstruct a continuous field out of the velocity data. Chapter 4 presents the resulting 3D velocity field and discusses possible findings and caveats of the result, before Chapter 5 summarizes them.

2. Taurus Molecular Cloud

2.1. Structure and Properties

Zucker et al. [37] traced the motions of nearby young stars back in time to reconstruct the origin of the Local Bubble: about 15 Myr ago, massive stars of the Scorpius-Centaurus (Sco-Cen) OB association were born near what is now the center of the Local Bubble, and their supernovae blew out the cavity. Sco-Cen has since drifted away from the center, while the shell has kept expanding and sweeping up the surrounding interstellar medium (ISM) into dense structures that can subsequently collapse and form new stars. Several nearby star-forming regions are attributed to this process, Taurus among them.

Taurus is embedded in a network of such expanding structures. It lies at the boundary between the Local Bubble and the Perseus-Taurus (Per-Tau) shell, a roughly spherical shell of swept-up gas and dust that connects Taurus to the neighboring Perseus molecular cloud [7]. Soler et al. [34] argue that the converging flows driven by these two structures triggered the formation of the cloud. A further large cavity lies nearby: the Orion-Eridanus superbubble, powered by the Orion OB1 association, extends into the broader vicinity of Taurus [29], placing the region within reach of several large-scale structures and making it a promising source of information about supernova-driven star formation in the solar neighborhood.

An interactive model presented by O'Neill et al. [30] depicts the Local Bubble not as a single, fully enclosed shell, but as a system of several shells that remain open to the Galactic halo; it is available online (interactive), and a snapshot is shown in Figure 2.1.

The TMC is one of the most intensively studied low-mass star-forming regions in the Milky Way. It spans a total of about 15×15 degrees on the sky and lies at a distance of approximately 140 parsecs from the Sun, although significant depth effects are present along the line of sight: the individual clouds range from 126.6 ± 1.7 to 162.7 ± 0.8 pc. The complex hosts a few hundred YSOs that are not randomly distributed but instead clustered into small groups [19]. Figure 2.2 shows the dust emission of the complex as seen on the sky.

The most recent census, based on Gaia DR3, revised the total membership to 532 adopted members distributed across 13 kinematic groups, most with median ages of 1–3 Myr, while neighboring associations have ages ranging from 13 to 56 Myr. These ages are estimated from the offset of each group's low-mass sequence in the Hertzsprung–Russell diagram relative to a reference population of assumed age, not from theoretical isochrones as in the catalog described later in Section 3.3. Samples of older stars (≥ 10 Myr) in the vicinity of Taurus were proposed to also be associated with the cloud but were not included in the catalog because of inconsistent kinematics compared to Taurus [26].

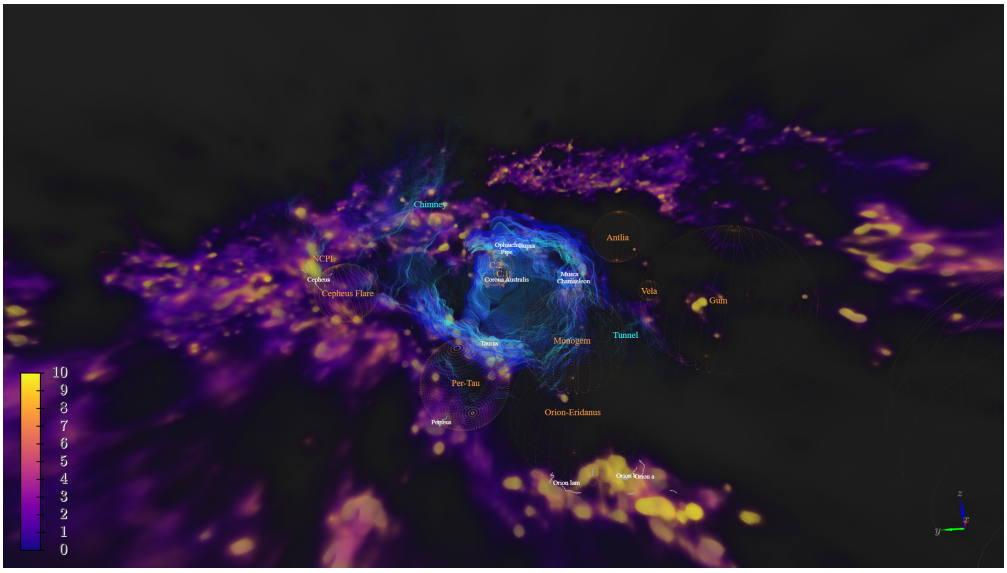


Figure 2.1.: Snapshot of the interactive three-dimensional dust map of the solar neighborhood from O'Neill et al. [30], showing prominent nearby molecular clouds, star-forming regions and further shell models like the Per-Tau shell around the low-density cavity of the Local Bubble. Taurus lies along its edge, adjacent to Perseus.

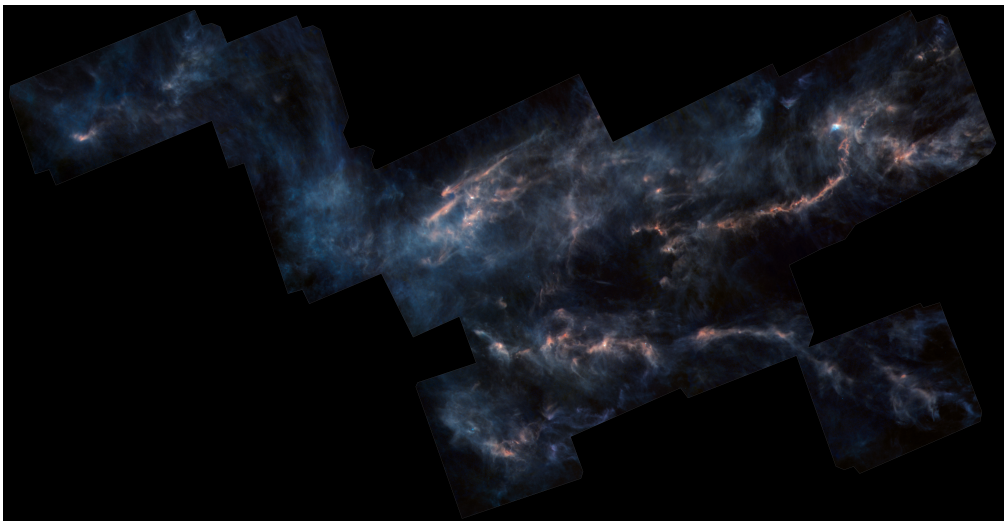


Figure 2.2.: "This mosaic combines several observations of the Taurus Molecular Cloud performed by ESA's Herschel observatory." [14]

2.2. Kinematics

The kinematics of Taurus have been the subject of several studies in recent years, enabled by increasingly precise astrometric data. An early investigation by Bertout and Genova [6] used proper motions supplemented by radial velocities to identify 94 probable members of the Taurus-Auriga moving group and to derive individual kinematic parallaxes, yielding a mean distance of 140 pc and a spatial velocity dispersion of approximately 6 $\mathrm{km\,s^{-1}}$ across the association.

A more complete and precise picture was established by Galli et al. [19], who combined Gaia DR2 astrometry with VLBI measurements from the GOBELINS project for a sample of 415 stars. Applying a hierarchical clustering algorithm, they identified 21 clusters, of which 13 could be associated with distinct molecular clouds, and derived their distances and spatial velocities. They found inter-cloud relative motions ranging from about 1 to 5 $\mathrm{km\,s^{-1}}$, a median inter-cloud separation of 25 pc, and a one-dimensional velocity dispersion among the clouds of about 2–3 $\mathrm{km\,s^{-1}}$. No significant expansion or contraction pattern was detected for the complex as a whole, although evidence for a potential global rotation was reported. The stellar velocities were furthermore found to closely match those of the underlying ^{13}CO molecular gas, confirming a tight dynamical coupling between the stars and the gas.

This picture was extended by Soler et al. [34], who combined 3D dust reconstruction with CO gas emission and neutral hydrogen observations to study the large-scale cloud dynamics. Their analysis revealed a velocity gradient across the cloud consistent with external compression, in line with the bubble geometry described above. Despite this progress, the internal kinematics of Taurus remain incompletely characterized, in part because radial velocity measurements are available for only a subset of its members. This motivates a complementary approach that uses the full available data of Gaia proper motions, as pursued in this thesis.

3. Methods

3.1. Information Field Theory

Rather than analyzing individual stars or predefined subgroups separately, this thesis combines the incomplete velocity information of the whole catalog into one continuous, three-dimensional velocity field that describes the motion of Taurus.

This is a common problem in physics: a continuous physical quantity, here a velocity field, must be reconstructed from a finite, noisy set of measurements. Information field theory (IFT) is a Bayesian framework designed specifically for this kind of problem [12, 13]. Instead of treating the unknown field as a small set of parameters, such as the coefficients of a simple analytic function, IFT treats it as the object to be inferred directly. The field, denoted s, can take on any value at any point in space and can therefore be seen as an infinite-dimensional random variable. Its shape is constrained purely by the data d and a set of prior assumptions.

Formally, the quantity of interest is the posterior probability $\mathcal{P}(s \mid d)$ of the field given the data. Bayes' theorem relates this posterior to the likelihood $\mathcal{P}(d \mid s)$, which describes how the data are generated from a given field configuration, and to the prior $\mathcal{P}(s)$, which encodes expectations about the field before any data are taken into account:

$$\mathcal{P}(s \mid d) = \frac{\mathcal{P}(d \mid s)\,\mathcal{P}(s)}{\mathcal{P}(d)}\,. \tag{3.1}$$

The evidence $\mathcal{P}(d)$ acts purely as a normalization constant and does not depend on the field s. For the inference of s it is therefore irrelevant and, as is common in many Bayesian approaches, never has to be evaluated explicitly; only the shape of the posterior as a function of s matters. Rather than a single field, the posterior gives a probability for every field that could have produced the data, and its scatter is the uncertainty of the reconstruction [12, 13].

In practice, writing the posterior in Eq. (3.1) down analytically requires assumptions this model does not make, such as a fixed correlation structure for the field. IFT provides the mathematical machinery to approximate this posterior numerically, and to do so in a way that also returns uncertainty estimates for the reconstructed field, not just a single best-fit map.

3.2. NIFTy - Numerical Information Field Theory

The numerical implementation of IFT used in this thesis is `NIFTy` [`NIFTy.re`; 11], a Python package built on `JAX` that extends the original `NIFTy` framework [3] with building

blocks for correlated field models and for the variational inference algorithms needed to approximate the posterior described before.

An important prior assumption of this model is that the velocity field is spatially correlated: nearby points are expected to have similar velocities, while the field can vary more or less smoothly depending on the physical scale considered. This kind of spatially correlated field follows and adapts the approach of Hutschenreuter et al. [22], who used the same underlying model and code to reconstruct the three-dimensional velocity field of the Sco-Cen OB association. Given the modest size of the Taurus sample of a few hundred stars, compared to over ten thousand for Sco-Cen, the model is kept deliberately simple with each velocity component being described by a single correlated field, without attempting to separate the region into several kinematically distinct populations, as was done for Sco-Cen, since the data would likely not constrain such a model.

The three-dimensional velocity field is modeled on a regular grid of $42 \times 23 \times 27$ cells of 3 pc, spanning a volume of $126 \times 69 \times 81$ pc around Taurus. Representing the field on this grid makes the inference computationally manageable, replacing an infinite number of parameters with a finite, though still large, set of values. Each velocity component is represented as a `NIFTy` correlated field, separated into a constant offset s_0, corresponding to the expected bulk motion of Taurus as a whole, and a spatially correlated deviation $\delta(\mathbf{x})$ from that offset,

$$s(\mathbf{x}) = s_0 + \delta(\mathbf{x})\,. \tag{3.2}$$

Rather than being built up cell by cell, the deviation is decomposed into Fourier modes, plane waves of a given wavelength and direction from which any field on the grid can be assembled. Short wavelengths carry the small-scale structure and long ones the large-scale variation. In this representation the deviation is generated as

$$\delta(\mathbf{x}) = \mathrm{FT}^{-1}\left[\sqrt{P(k)}\,\xi(\mathbf{k})\right](\mathbf{x})\,, \tag{3.3}$$

where the $\xi(\mathbf{k})$ are independent random numbers drawn from a standard normal distribution, one for each mode. They carry no structure themselves; it is the power spectrum $P(k)$ that assigns each mode its typical amplitude on the spatial scale k^{-1} and thereby fixes the correlation structure of the field. The inverse Fourier transform FT^{-1} then adds the weighted modes back together into a field in real space. The inference only has to find the ξ that reproduce the data instead of a correlated value in every grid cell. $P(k)$ itself is not fixed by hand: its amplitude and slope are inferred from the data, so that the balance between small-scale and large-scale structure is set by the observations rather than assumed a priori.

The likelihood does not compare the field to space velocities, but to the measured quantities. For each star the field is evaluated at its position through interpolation (further discussed in Chapter 4) and projected onto the two proper-motion components on the sky and, where available, the radial velocity along the line of sight, with additive Gaussian measurement noise. A star without a radial velocity therefore constrains the field only within the plane of the sky, but since the prior correlates neighboring positions, its missing component is supplied by the stars around it. This is what allows the full sample to be used instead of only the part of it with complete velocities.

The nominal Gaia uncertainties do not necessarily capture the full observational error, for example because unmodeled stellar peculiar motions or unresolved binary systems add scatter beyond the reported values. The model can therefore rescale the noise level of the radial velocities by an η parameter per star that is estimated from the data. This data-driven noise estimation is applied only to the radial-velocity measurements. The proper motions are kept at their reported Gaia uncertainties. Its usage is discussed in Chapter 4, where its effects on the reconstruction are examined.

Given this likelihood and the correlated-field priors, the posterior distribution over the velocity field is then approximated using `NIFTy`'s variational inference scheme, geometric variational inference (geoVI), and metric Gaussian variational inference (MGVI) [23, 16]. Both approximate the posterior by a simpler distribution $\mathcal{Q}(s)$, fit by minimizing the Kullback–Leibler divergence

$$D_{\mathrm{KL}}(\mathcal{Q} \,\|\, \mathcal{P}) = \int \mathcal{D}s \; \mathcal{Q}(s) \; \ln \frac{\mathcal{Q}(s)}{\mathcal{P}(s \mid d)} \,, \tag{3.4}$$

which is zero only if $\mathcal{Q}$ exactly matches the true posterior. Expressing the posterior through Bayes' theorem (Eq. (3.1)) reduces the objective to quantities the model provides explicitly, the likelihood and the prior, with the evidence $\mathcal{P}(d)$ remaining merely as an additive constant.

In practice, the algorithm iteratively refines this approximation by alternating between updating the most probable field configuration and drawing an increasing number of posterior samples around it, in order to capture the local shape of the posterior more accurately as the fit converges. The final result of this procedure is a set of posterior samples of the full 3D velocity field, from which both a mean velocity map and a corresponding uncertainty (standard deviation) map are computed at every grid point, providing the reconstructed velocity field of Taurus together with an estimate of its reliability [23, 16].

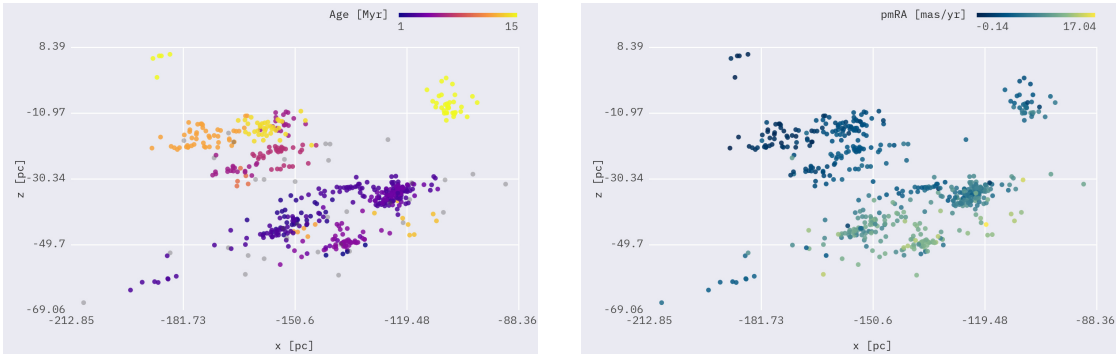
3.3. Dataset

The analysis is based on a catalog of 557 stars associated with the TMC, identified with the `SigMA` clustering algorithm of Ratzenböck et al. [32] applied to Gaia DR3 astrometry [18]. The same membership catalog underlies the analysis of Alves et al. [1]. Members that also appear in the census of Luhman [26] are flagged as such. For each star the catalog provides the full astrometric solution: equatorial coordinates (α, δ), trigonometric parallax ϖ and proper motions in right ascension and declination ($\mu_{\alpha*}$, μ_δ) together with radial velocities, where available, all with their respective uncertainties. The renormalized unit weight error (RUWE) serves as a quality indicator for the astrometric solution, quantifying how well the single-star model fits the data, with values close to one indicating a good fit [31]. All 557 stars possess proper motion measurements, while radial velocities are available for only 180 of them, representing roughly one third of the sample (Figure 3.4). Photometric data are provided in the form of the Gaia G-band magnitude and the colors $G - G_{\mathrm{RP}}$ and $G_{\mathrm{BP}} - G_{\mathrm{RP}}$. Stellar ages, visual extinctions A_V, and masses are included for a subset of the sample. They are obtained with the `Chronos` algorithm [33],

which fits theoretical isochrones to the Gaia color–magnitude diagram of each cluster while solving for extinction simultaneously, so that every member of a cluster is assigned the same age.

The three-dimensional heliocentric Galactic Cartesian positions (X, Y, Z) and corresponding velocity components (U, V, W) are computed from the astrometric and radial-velocity data, with the X-axis pointing toward the Galactic center, Y in the direction of Galactic rotation, and Z toward the Galactic north pole.

Before any field reconstruction, several properties of the data can already be analyzed directly. Figure 3.1a shows the stellar positions projected onto the X–Z plane, colored by stellar age. The members are not uniformly distributed but fall into several spatially separated groups that also differ in age, with a visible age gradient between the two large subgroups and a small, spatially distinct group of older stars (15 Myr). Considering the velocity data, the proper motion in right ascension reveals the same division, so that ages and velocities appear correlated (Figure 3.1b).



(a) Stellar ages

(b) Right ascension proper motion

Figure 3.1.: Positions of the catalog stars projected onto the X–Z plane, colored by stellar age (left) and right ascension proper motion data (right). Stars shown in grey have no data available. Several spatially separated groups are visible, indicating that star formation in Taurus is both clustered and spread out in age.

The population spans ages from 1 to 15 Myr, a wider range than the median ages of 1–3 Myr quoted in Section 2. This is mainly a question of membership: the clustering used here also recovers comoving groups of ∼12–15 Myr that Luhman [26] assigns to neighboring associations rather than to Taurus itself. The ages in the two catalogs are moreover determined with different methods. The bulk of the sample is nonetheless young with most of the stars between 1 and 7 Myr, and the distribution peaks at 3 Myr, in agreement with the median group ages of 1–3 Myr reported by Luhman [26]. The age distribution of the catalog is shown in Appendix A.1.

Additionally, Figure 3.2 shows a color-magnitude diagram (CMD) of the catalog, colored by age to highlight the different subgroups, with isochrones for 1, 3, and 15 Myr overlaid [5]. One star without available photometric data is excluded from the diagram.

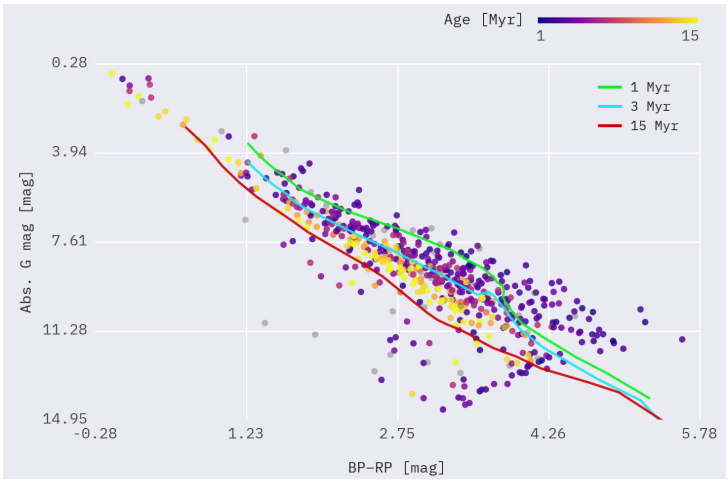
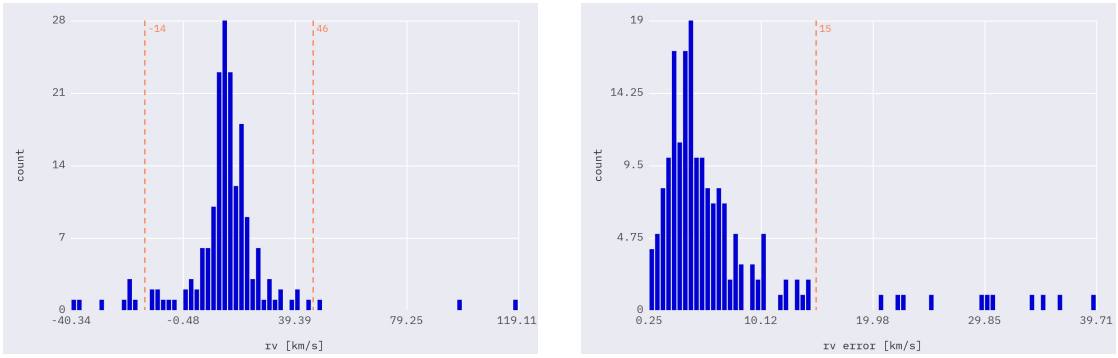


Figure 3.2.: CMD of the Taurus complex colored by the ages of the stars. Stars shown in grey have no age data available. Isochrones of ages 1, 3 and 15 Myr are overlaid.

The distribution of the measured radial velocities and their uncertainties gives a first indication of the data quality. Figure 3.3 shows a histogram of all available radial velocities alongside the distribution of their reported uncertainties. The bulk of the measurements clusters around the systemic velocity of Taurus, while a small number of clear outliers lie well outside the plausible range, and a fraction of the measurements carry large uncertainties. Both features motivate the filtering described below.



(a) Radial velocities.

(b) Radial-velocity uncertainties.

Figure 3.3.: Distribution of the measured radial velocities (left) and their reported uncertainties (right) for the 180 stars with available radial velocities. Quality cuts are indicated.

Before constructing the field, the sample is filtered to remove unreliable measurements. The radial velocity data are distributed as a Gaussian curve with approximate mean and standard deviation of 16 and 6 $\mathrm{km\,s^{-1}}$ respectively. A 5σ cut was chosen to ex-

clude outliers from this distribution, which leaves radial velocity values in the range $[-14, 46]\,\mathrm{km\,s^{-1}}$. Furthermore, stars with uncertainties on their RV measurement greater than $15\,\mathrm{km\,s^{-1}}$ are excluded from the likelihood as this marks the border between the bulk of measurements and the sparse tail of poorly constrained values. Stars with RUWE > 2.5 are additionally excluded, as elevated RUWE values are indicative of unresolved binary systems or a poor astrometric solution that may bias the velocity measurement [24, 31]. This value was chosen according to Fitton et al. [15], who argue for a more conservative binary threshold to account for the presence of disk material near young stars that could influence the single-star solution. These cuts are applied only to the radial-velocity measurements, since proper motions carry substantially smaller uncertainties and are less strongly biased by unresolved binaries, which can significantly affect individual radial-velocity values [10]. Due to the limited number of stars with radial-velocity measurements, these cuts are set deliberately lenient in order to retain as many data points as possible, excluding only the most severe outliers.

The data entering the inference contain all 557 proper motion measurements and 123 radial velocity values left after the quality cuts. Figure 3.4 shows a three-dimensional representation of the stellar data, with the stars whose radial velocities entered the inference marked separately. The purpose of this visualization is to illustrate how few radial velocities remain usable: after the quality cuts their share of the catalog drops from one third to less than a quarter. Vector-based illustrations of the three-dimensional velocities are provided in Appendix A.2.

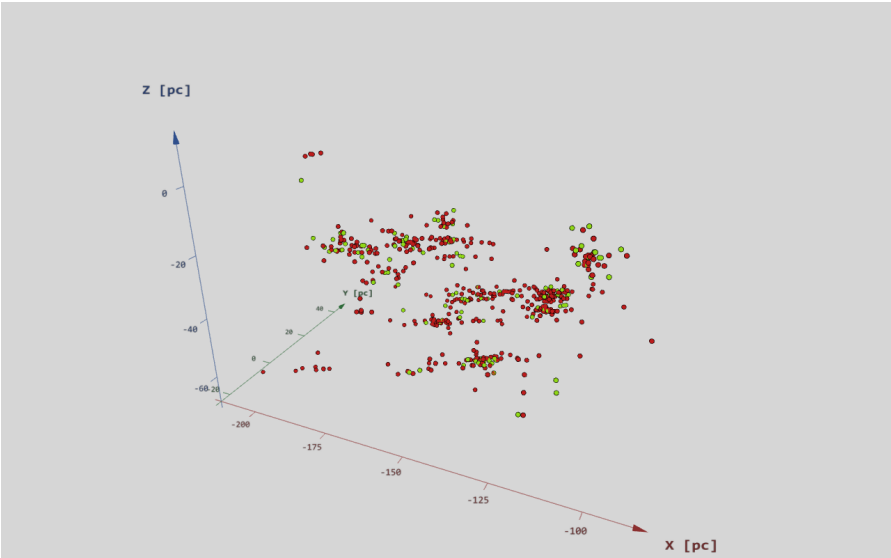


Figure 3.4.: Three-dimensional heliocentric Galactic Cartesian positions of the 557 catalog stars. Stars for which a radial-velocity measurement was used in the analysis (123 stars) are highlighted in green.

4. Results and Discussion

4.1. The Reconstructed Velocity Field

Firstly, a short clarification on the following results: given the small sample of stars that the catalog of the TMC consists of and the limited RV data, it was challenging to produce a satisfactory velocity field. The appropriate selection of prior parameters, as well as trying to constrain the field not to diverge into configurations that are inconsistent with the data, proved difficult. It took many tries to arrive at somewhat promising results, and even then it was another challenge to reproduce such results. Taurus simply has too few data points for the algorithm to reliably constrain the field, smooth out the noisy data and deal with outlier stars at the same time.

For this reason, this section does not present and discuss the following reconstruction of the field as the final and correct result of this thesis, but rather as a work in progress and the most promising result acquired at this point. It is therefore also not overly beneficial to analyze the field in detail; it is instead discussed and reviewed qualitatively. Section 4.2.1 discusses different runs of the algorithm with different parameters or different quality cuts, and even runs with the same parameters where the posterior yielded different results. Features that appear consistently across these runs can be read as actual data driven properties rather than artefacts resulting from the choice of run.

Apart from the data cuts, the model itself was revised several times over the course of this thesis. Most of these changes came out of a series of unsuccessful iterations, each giving further insight into the appropriate choice of model and parameters.

First, the η noise estimation was, after considerable deliberation, removed from the final model. It gives the inference the freedom to downweight the already small sample of radial velocities, whose uncertainties are far larger than those of the proper motions. Since the field is constrained mainly by the far more precise proper motions, the noise estimation ends up disregarding the radial velocities altogether, inflating their uncertainties to the point where the measurements no longer constrain the field. With a sample as small as that of Taurus, this effect is not compensated by a large number of remaining measurements, as it would be in a richer region.

Second, the readout of the field at the stellar positions was changed in the final iterations. Previously, each star was compared to the grid point at the same corner of the voxel containing it, so that the point of evaluation could be displaced from the star by up to one cell, or 3 pc, along every axis. The new readout is a trilinear interpolation: the field is evaluated at all eight corners of the surrounding voxel and weighted by their proximity to the star. This removes the spatial offset between a star and the point at which the field is evaluated. The readout now varies continuously with position, so stars

that share a voxel are no longer compared to one and the same grid value, and their residuals depend on where in the cell they sit rather than on the grid spacing.

With that said, Figure 4.1 presents the resulting velocity field. Vectors are drawn at every grid cell, with a mass-density mask hiding the regions that are not covered by data (Section A.3). There the reconstruction is no longer constrained by the likelihood and merely extrapolates the prior. The same mask is applied to every subsequent figure. The general direction of the bulk motion in Figure A.2a remains, but with smaller magnitudes than in the data as `NIFTy` smooths the field and filters outliers. The main complex of Taurus appears to move in the same direction while different parts of the cloud move at different velocities revealing subpopulations.

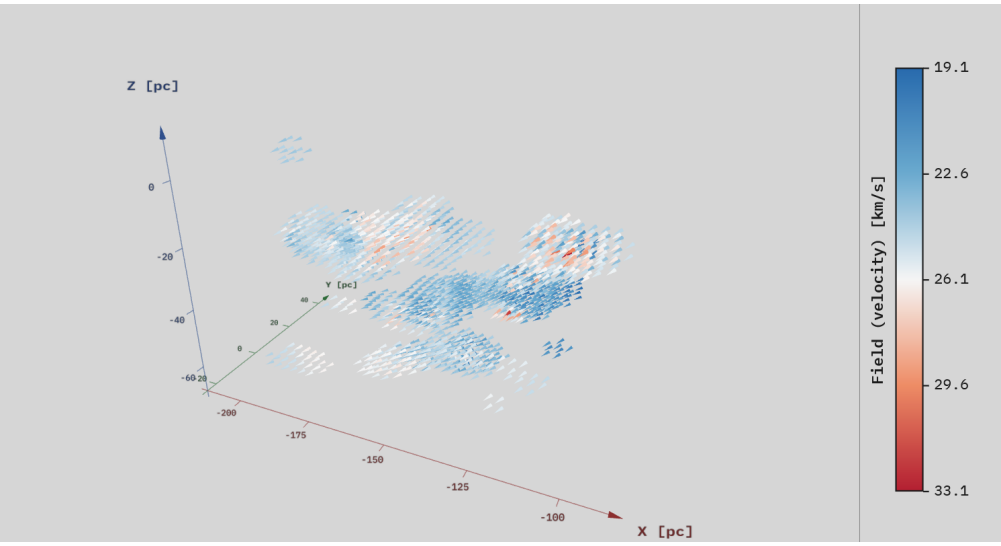


Figure 4.1.: Three-dimensional representation of the velocity field in the solar barycentric reference frame represented as vectors.

Several small-scale peaks stand out against the otherwise smooth field. They appear throughout the region, also where stars are plentiful, since the roughly 550 members are still outnumbered by the grid cells they have to constrain: a star whose velocity deviates from its neighbors drives a local peak rather than being averaged down. Whether such a feature traces a real motion or a single deviant measurement is decided by its reappearance across runs, which is examined in Section 4.2.1.

The peaks also stretch the colorbar, masking the smaller differences across the rest of the region into a narrow band of color. Figure 4.2 therefore repeats the field with the colorbar clamped to 21–28 $\mathrm{km\,s^{-1}}$, where the cells outside this range are drawn in the limiting colors. The limits sit roughly symmetrically around the bulk of the distribution, so that only the peaks are clipped.

It should be noted that Gaia measures in the barycentric reference frame [18], so the velocities describe how Taurus moves relative to the Sun. The bulk motion of the field therefore says as much about the motion of the Sun as about that of Taurus, whereas

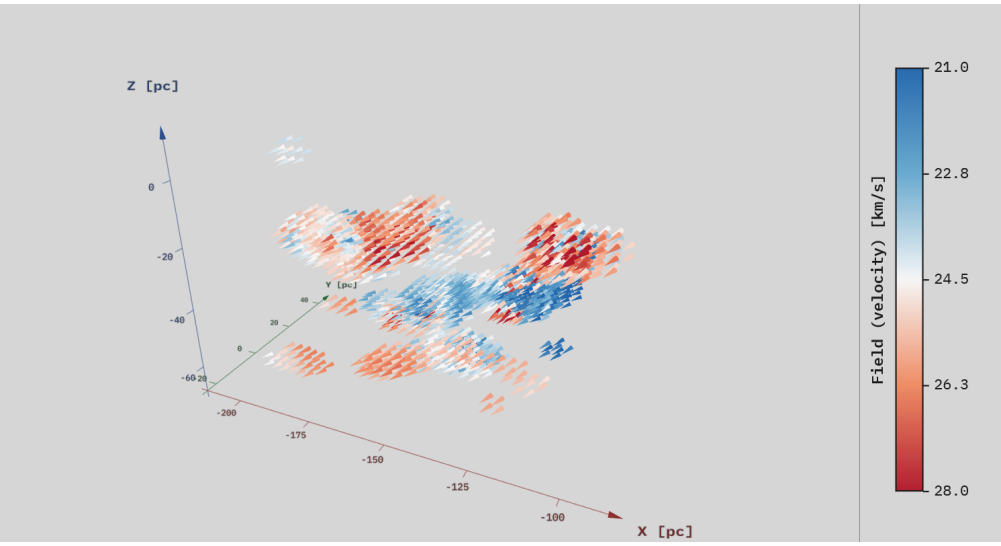


Figure 4.2.: The velocity field of Figure 4.1 with the colorbar clamped to 21–28 km s^{-1}, showing the variation between the subpopulations that the small-scale peaks would otherwise hide.

the relative motions within the region are unaffected by this choice of frame. To gain physically relevant information about Taurus, a change of reference frame would be required as performed later in Section 4.1.1.

Comparing the velocity magnitudes to the stellar ages in Figure 4.3 reveals a systematic trend between the two quantities: the youngest stars (1–5 Myr) reach smaller velocity magnitudes than the older subgroups (6–15 Myr), forming an age–velocity gradient across the region. This is not an artefact of this particular reconstruction, as the same property appears in most other runs of the algorithm and is already present in the stellar data itself (Figure 3.1). As in Figure 4.2, the colorbar is clipped so that the small-scale peaks do not compress the scale of the field.

The comparison is, however, tied to the selection of reference frame. Subtracting a constant offset, as done in Section 4.1.1, changes both the magnitude and the direction of every velocity vector, and the ordering of magnitudes is not necessarily preserved under such a shift.

A 3D viewer, built for the analysis of the resulting vector field and the stellar data, is available at vectorfield viewer.[1] Most plots shown here result from this application.

4.1.1. Taurus Reference Frame

Choosing an appropriate reference frame is essential for extracting physical information about Taurus and for any further analysis of momentum or energy. An important choice is the frame internal to the complex, in which the motion of Taurus as a whole is re-

[1] https://dereppi.github.io/Viewer/

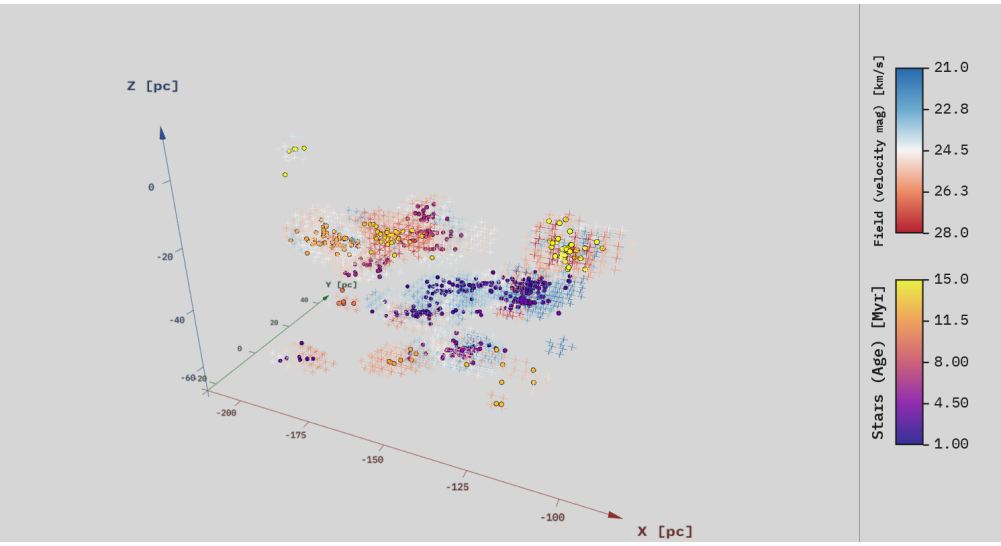


Figure 4.3.: Comparison between star ages (dots) and velocity magnitudes overlaid.

moved. It is obtained by subtracting the mean field velocity, weighted by the mass density from Section A.3. Since the reconstruction is heliocentric, this mean contains the motion of Taurus itself together with relative solar motion. In this run it evaluates to $(-17.22, -14.48, -9.25)\ \mathrm{km\,s^{-1}}$ in galactic coordinates, corresponding to a bulk speed of $24.32\ \mathrm{km\,s^{-1}}$. Within a single run the posterior uncertainty of this mean is small, about $0.4\ \mathrm{km\,s^{-1}}$ in U and below $0.1\ \mathrm{km\,s^{-1}}$ in V and W, but between runs the values scatter by several $\mathrm{km\,s^{-1}}$ in every component. The mean motion is therefore not a reliable measurement of the motion of Taurus, and the internal frame defined by it inherits that limitation. For comparison, Galli et al. [19] derive $(-16.2, -12.8, -8.9)\ \mathrm{km\,s^{-1}}$, or $22.5\ \mathrm{km\,s^{-1}}$ in magnitude, from 92 members with radial velocities. The same values enter the reconstruction as the prior mean of the velocity offset, so the agreement is not an independent confirmation. The prior is broad enough, with a width of the order of $10\ \mathrm{km\,s^{-1}}$, to let the data move the offset away from it, but part of the recovered mean still reflects that choice. Figure 4.4 shows the velocity field in the discussed reference frame.

With the common bulk motion removed, the differences between the subgroups stand out: which parts of the complex move faster than the region as a whole, and in which direction they depart from it. Since an error in the subtracted mean shifts every vector by the same amount, groups can be compared to each other within one run, while the absolute magnitudes and directions shown here carry the uncertainty of the frame itself.

This reference frame also enables the analysis of internal energy distributions using the kinetic energy of the stars $E = \frac{1}{2}mv^2$. Observing where high amounts of energy are stored could reveal information about the origin of Taurus. As the energy analysis depends on the correctness of this reference frame, a thorough investigation is not pursued here. However, integrating $\frac{1}{2}\rho\,|\vec{v}|^2$ over the reconstruction volume, with ρ the stellar mass

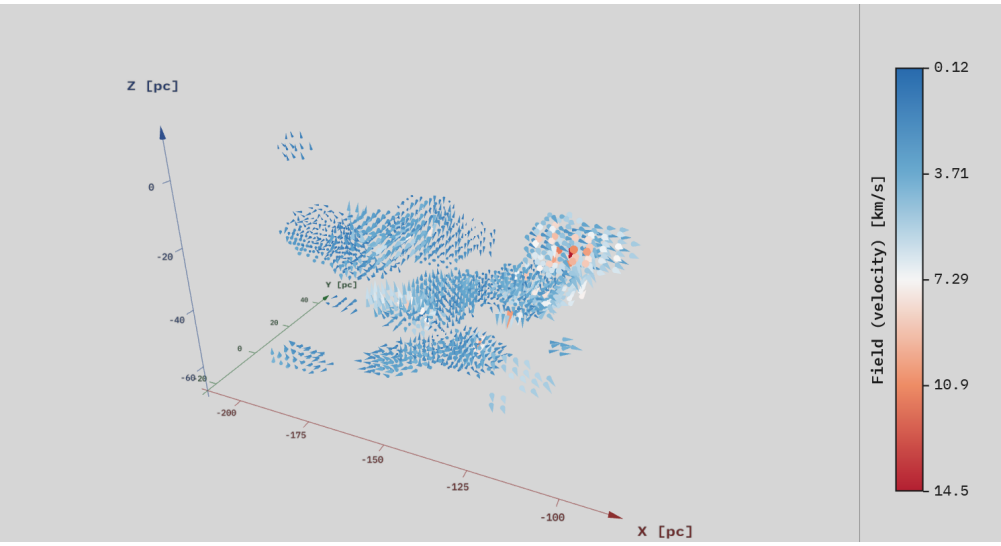


Figure 4.4.: The velocity field in the Taurus internal reference frame with the bulk motion of the complex subtracted.

density of Section A.3, gives a total kinetic energy of $2.8 \cdot 10^{39}$ J in this particular run.

4.1.2. Derivatives

Two additional quantities derived from the field describe its local behavior: the divergence $\nabla \cdot \vec{v}$, positive where the field expands and negative where it contracts, and the vorticity $\nabla \times \vec{v}$, which describes local rotation and shear.

These two quantities have the advantage of being independent of the reference frame, since subtracting a constant leaves the derivatives unchanged. They are thus not affected by the problem of the previous section. What limits them is resolution: a derivative responds to the smallest scales of the reconstruction, which are the least constrained by the data and also the ones that differ most between runs (Section 4.2.1). They are computed from the mean field rather than from the individual posterior samples, so no uncertainty is attached to them.

The maps, shown in Appendix A.4, are therefore only read qualitatively. Galli et al. [19] find no significant expansion or contraction of the complex, quoting an expansion velocity of 0.0 ± 0.1 km s^{-1}, while they do report a potential global rotation of 1.5 ± 0.1 km s^{-1}.

4.2. Validation

`NIFTy` provides a set of posterior samples which all represent a complete field; from these samples, a mean and uncertainty (standard deviation) field is created. The uncertainty estimation is dependent on the current posterior. If the algorithm is stuck in a local minimum where the posterior varies relatively little, the uncertainty is then estimated

also on this local minimum, whereas the global optimum could still lie outside of this uncertainty. It should therefore not be interpreted as deviation between runs but rather as indicator of how well the field is constrained at this point and how much it varies. The posterior uncertainty of the run discussed above is shown in Figure 4.5.

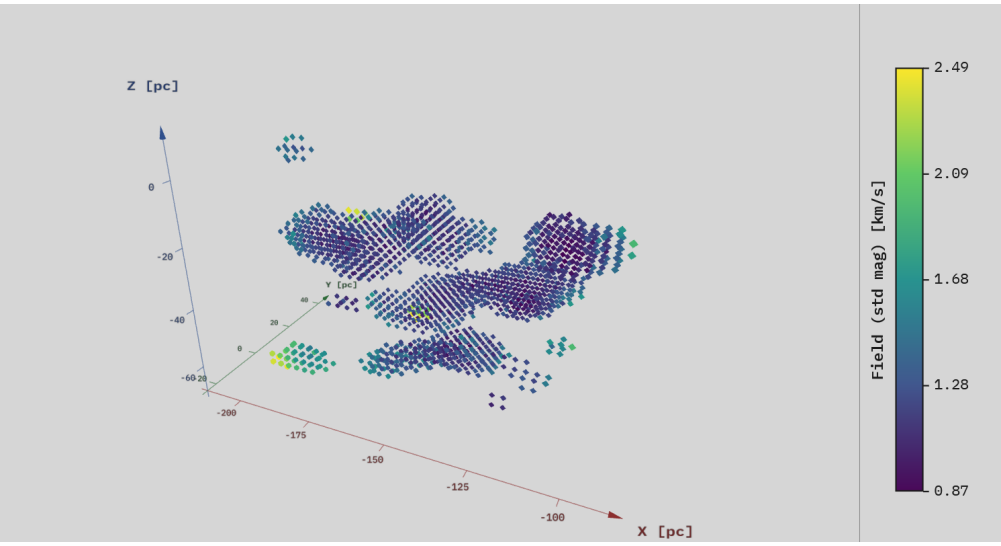


Figure 4.5.: The posterior uncertainty represented as magnitude of all three components.

A key insight here is that while the posterior is well constrained in the central regions, where most of the data lie, the edges have much higher uncertainties. Another finding is that the uncertainty is dominated by the U component with a median uncertainty of 0.93 $\mathrm{km\,s^{-1}}$ while the V and W components have median uncertainties of 0.46 and 0.57 $\mathrm{km\,s^{-1}}$ respectively. This comes from the specific property of the TMC that U points almost along the line of sight and consists mostly of the radial velocity component, which has far greater measurement uncertainties than the proper motions and in turn contributes more to the variation of the posterior samples.

The residuals between the data and the field values provide another quality check on the posterior. Stars carry peculiar motions that a continuous field cannot represent, and fitting every star exactly is neither possible at this resolution nor desirable, so these residuals are expected to be nonzero. Since the plain differences between stellar and field values carry only limited information about the accuracy of the fit, the weighted residuals are more informative: each absolute difference is divided by the effective uncertainty of the star. A weighted residual of one or below indicates that the field matches the star within its uncertainty. The residual statistics are presented in Table 4.1.

The most noticeable feature of this statistic is that the proper motions are clearly not fit within their uncertainties. The median fit lies 4.71 and 6.83 standard deviations away from the actual value, while the median radial velocity fit lies within almost exactly one σ. The fit is poor by this measure, but the reason is straightforward. The effective uncertainty of the stars in this model consists purely of the measurement errors. The

Table 4.1.: Residuals between the stellar data and the posterior mean field. The left block lists the absolute differences, the right block the same differences divided by the effective uncertainty σ_{eff} of the respective star. N is the number of stars entering each row.

		Residual			Weighted residual [σ_{eff}]		
Component	N	Mean	Median	Max	Mean	Median	Max
$\mu_{\alpha*}$ [mas yr^{-1}]	557	0.928	0.393	13.905	10.29	4.71	240.48
μ_{δ} [mas yr^{-1}]	557	1.126	0.434	21.159	16.31	6.83	330.28
v_r [km s^{-1}]	123	5.737	3.550	28.845	1.59	0.98	31.99

median error is 0.06 mas yr^{-1} for $\mu_{\alpha*}$ and 0.04 mas yr^{-1} for μ_{δ}, with median measurements of about 7 and -21 mas yr^{-1} respectively. So the algorithm tries to accurately fit data with an uncertainty of about one percent of the measurement or less. The stars, however, do not move as a continuous field; they possess peculiar motions of their own. Galli et al. [19] determined the three-dimensional velocity dispersion inside Taurus to be $\sigma = 2.7$ km s^{-1} and describe the internal motions in the radial direction of the complex as "dominated by random motions rather than an organized expansion or contraction pattern". This suggests a change to the model that accounts for these peculiar velocities.

4.2.1. Comparison

This section serves as a case study of different iterations of the algorithm, comparing runs that differ in their parameters, in the applied quality cuts, or merely in their results. Some of these runs show which features of the reconstruction survive such a change and can be read as data driven, while others expose the weaknesses of the model that motivate the extension discussed in Section 4.2.2.

These runs are also available at vectorfield viewer with the respective names of the runs mentioned here. To keep this work clear from yet another set of large three-dimensional plots, the reader is advised to confirm the described features in the interactive viewer, where each run can be inspected freely and its exact parameters are listed alongside the reconstruction.

The runs were not all produced with the same code. Over the course of this thesis several errors in the pipeline were fixed, and the readout of the field at the stellar positions was replaced by the trilinear interpolation described at the beginning of this chapter. A difference between two runs can therefore originate from the parameters, from the quality cuts, or from the state of the pipeline they were run with.

The earliest iteration considered here is `parameters_old`. It combines loose priors, which leave the field free to vary strongly and on small scales, with quality cuts that discard more data than necessary: only 89 stars enter the fit with a radial velocity, where the later runs keep between 110 and 160. Its noise estimation then behaves as described above: it inflates the radial-velocity errors by more than a factor of two, leaving the field

to be shaped almost entirely by the proper motions. The result is the "split" of the field that appears in most runs with an active noise estimation, where the vectors diverge in opposite directions along U and separate the cloud into two halves. The data show no such pattern. As described in Section 4.2, U is the component the radial velocities constrain; once they are downweighted, the algorithm is free to choose its direction. A strong small-scale peak in the region of the younger stars of $\sim 80\ \mathrm{km\,s^{-1}}$ appears alongside it. Overall, this iteration yields no physical insight into the motions of Taurus. Its value lies in what followed from it: the noise estimation was dropped eventually, and the quality cuts were revised to keep more of the few radial velocities that are available.

Another interesting run is `ruwe_test`. Its more lenient quality cuts leave 112 radial velocities, and a steeper prior on the slope of the power spectrum should reduce the overfitting on small scales. It is the exception among the runs with an active noise estimation, as it does not develop the patterns described above. The median η stays near 2, so the radial-velocity errors are inflated by some 40 per cent instead of a factor of two or more, and the measurements keep constraining the field. Accordingly, the split along U is absent. Why the noise estimation settles at this level here and not in the other runs cannot be read off the run itself. The field is dominated by large-scale structure and carries few small-scale fluctuations, in line with the steeper slope prior. The age–velocity gradient described above nevertheless reappears, which supports reading it as a property of the data rather than of one particular reconstruction.

A last example of these iterations, `test_new_param`, is the configuration the rewritten pipeline started from. The noise estimation is switched off, the quality cuts are the loosest of all runs discussed here and leave 156 radial velocities, and the prior on the fluctuations is tightened so that the field can no longer vary as freely as before. With the radial velocities at full weight, the split along U does not appear. The mean motion, however, comes out at $U \approx -8\ \mathrm{km\,s^{-1}}$, roughly half the value of Galli et al. [19] and of the runs discussed in Section 4.1.1, even though the offset prior is centered on the literature value. A small-scale peak appears here as well, but at a different position than in `parameters_old`: such features move with the run rather than marking a fixed location in the region. They persist because the field is still the only place where the deviation of an individual star can go, which is what the next section targets.

4.2.2. Peculiar motions

The demonstrations in Section 4.2.1 and the residuals above point at the same gap in the model: the uncertainty assigned to a star contains only its measurement error. For the proper motions that error is far smaller than the velocity scatter of the stars themselves, so the motion a star has on top of the large-scale flow enters the residuals in full and is judged against an uncertainty tens of times smaller, reaching several hundred σ in the extreme cases of Table 4.1. The correlated field prior limits how closely the field follows individual stars, but it cannot separate stars that share similar positions but move differently: a single field can only represent their average motion, and telling them apart would require a second field component [22]. With a sample of this size that route is not available (Section 3.2), so the deviation has to be accounted for on the side of the

noise instead.

A simple approach for this is an intrinsic scatter term [21, Sect. 8]. It starts from the observation that the offset of a star from a smooth field has two independent causes: the measurement error reported by Gaia, and the peculiar velocity the star has on top of the large-scale flow. The first is known for every star, the second is not. Fitting a peculiar velocity for each star separately would add hundreds of free parameters that the data cannot constrain, much like the η noise estimation discussed above. What can be captured by a single number, however, is how widely the stars scatter as a population: their peculiar velocities are assumed to be drawn from one common Gaussian distribution of width σ_{int}, taken to be the same in all directions.

The measurement errors are already modeled as Gaussian (Section 3.2), so both contributions are Gaussian and independent of one another: their variances add, and their sum is Gaussian again. A star is therefore expected to lie further from the field than its measurement error alone would suggest, by an amount set by σ_{int}, and the two enter the likelihood added in quadrature. Apart from this new parameter, nothing else at the model changes, each star still contributes a single residual between its measurement and the field, only now this residual is judged against the widened uncertainty. For the radial velocities the effective uncertainty is therefore

$$\sigma_{v_r,\mathrm{eff}}^2 = \sigma_{v_r}^2 + \sigma_{\mathrm{int}}^2 \,, \tag{4.1}$$

with σ_{int} in $\mathrm{km\,s^{-1}}$. For the proper motions σ_{int} first has to be converted to $\mathrm{mas\,yr^{-1}}$ at the distance d of the respective star,

$$\sigma_{\mu,\mathrm{eff}}^2 = \sigma_{\mu}^2 + \left(\frac{\sigma_{\mathrm{int}}}{\kappa\, d}\right)^2 , \qquad \kappa = 4.7405 \,, \tag{4.2}$$

with d in kpc. The field is then no longer required to pass through every star; it only has to reproduce the stellar velocities to within the width of their own velocity distribution, so that a deviation the field cannot represent no longer counts against the fit. However, since σ_{int} and the amplitude of the field are constrained by the same data and can be traded against one another, the same measurements are explained either by a structured field with little intrinsic scatter or by a smoother field with a wider one. Which of the two the inference settles on depends on the prior placed on σ_{int}, and a value chosen too generously can absorb velocity differences that belong in the field.

The scale of the parameter is set by the region itself, but the 2.7 $\mathrm{km\,s^{-1}}$ of Galli et al. [19] cannot be adopted directly: the value is three-dimensional and measures the scatter of all Taurus members around a single mean velocity. Most of it is large-scale — the clouds move relative to each other by 1 to 5 $\mathrm{km\,s^{-1}}$, and these coherent motions are what the reconstructed field describes on its own. σ_{int} only has to cover what the field cannot follow, the deviation of a star from its immediate surroundings on scales below the 3 pc of a grid cell, so the complex-wide value is an upper limit. Within the individual clouds Galli et al. [19] report dispersions of 0.2 to 1.6 $\mathrm{km\,s^{-1}}$ per component, which still contain the measurement errors.

The run `test_with_sigma` applies this to the setup of the default run, with a log-normal prior on σ_{int} of mean 0.8 $\mathrm{km\,s^{-1}}$, well inside the range argued for above. The

posterior settles at 0.99 $\mathrm{km\,s^{-1}}$, so the data ask for slightly more scatter than the prior suggests. Table 4.2 compares the weighted residuals with those of the default run. The absolute differences barely move, so the reconstruction is largely the same, but the median weighted residuals of the proper motions fall from 4.71 and 6.83 to 0.34 and 0.45, and the extreme cases from several hundred σ_{eff} to below ten. Both medians now lie below the 0.67 expected for correctly scaled uncertainties, while the radial velocities stay at 0.91: one common width cannot serve both at once.

Table 4.2.: Weighted residuals of the default run and of `test_with_sigma`, in units of the effective uncertainty σ_{eff} of the respective star. For `test_with_sigma` this uncertainty includes the inferred intrinsic dispersion of 0.99 $\mathrm{km\,s^{-1}}$, added in quadrature to the measurement error following Equations 4.1 and 4.2. N is the same for both runs: 557 stars for the proper motions and 123 for the radial velocities.

	Default run			`test_with_sigma`		
Component	Mean	Median	Max	Mean	Median	Max
$\mu_{\alpha*}$	10.29	4.71	240.48	0.57	0.34	5.31
μ_{δ}	16.31	6.83	330.28	0.70	0.45	8.18
v_r	1.59	0.98	31.99	1.25	0.91	7.08

Visually, the run again produces a field dominated by large scales, with no outstanding small-scale fluctuations. This is what the model is meant to do, as it infers the smooth field underlying the stellar motions rather than the motions themselves. One pattern stands out: large velocities in the U component on the far side of Taurus, where hardly any star with a measured radial velocity is left. Since U is the component the radial velocities constrain, the field is extrapolated there rather than measured, which makes this an edge effect rather than a property of the region. Continuing the inference for further iterations may also resolve it.

This run remains a test. The appropriate value of σ_{int}, and whether inferring it alongside the field is the right choice at all, would need further investigation; the prior used here was settled on in a few earlier iterations by comparing the resulting fits. It should also be kept in mind that widening the uncertainties improves the fit statistic without necessarily bringing the field closer to the data: the absolute differences are, if anything, slightly larger than before, and only the standard they are measured against has changed. As mentioned before, since the inference can trade σ_{int} against the amplitude of the field, the recovered 0.99 $\mathrm{km\,s^{-1}}$ should not be read as a measurement of the peculiar velocity dispersion in Taurus.

The most severe limitation, however, remains the data itself: Taurus holds few stars and even fewer with a reliable radial velocity. Compared with pre-Gaia times the sample is immensely better in both size and precision, but a reconstruction of this kind would profit from a revision once Gaia DR4 [17] provides a larger and better characterised set of radial velocities.

5. Conclusion

The velocity field reconstructed in this thesis reproduces the bulk motion of the Taurus complex and, once that motion is subtracted, separates its subgroups by their internal velocities. The age–velocity gradient already visible in the stellar data persists in the field and across most runs, and the divergence and vorticity maps add a qualitative picture that does not depend on the chosen reference frame. What the field supports are statements about which parts of the complex move differently from each other, rather than measurements of individual velocities: if the bulk motion is off, every vector of the field is off by the same amount, so the error cancels when two regions are compared but remains in any velocity value quoted on its own.

The caveats are as important as the result. The mean motion, and with it the internal reference frame, varies by several $\mathrm{km\,s^{-1}}$ between runs, which enters every absolute magnitude and direction in that frame, while a comparison between two regions of the same run is unaffected by that shift. Small-scale peaks appear throughout the region but at different positions in different runs, which marks them as consequences of individual stars rather than properties of the complex. The reconstruction also inherits the geometry of the region: U points almost along the line of sight and is therefore the component that the scarce radial velocities have to constrain, which makes it the least certain part of the field and the source of its most conspicuous artifacts. Underlying all of this is the central assumption that the stars follow a smooth field. They do so only approximately, and the model offers no place for the difference: a peculiar motion ends up either in the field or in the residuals. Granting the stars a common intrinsic dispersion repairs the residual statistics, but the value obtained this way is a fit parameter rather than a measurement, since the same data constrain it and the amplitude of the field.

Most of these limitations trace back to one source: too few stars, and radial velocities for only a fraction of them. With 557 members on a grid of about 26 000 cells and 123 usable radial velocities, the data cannot constrain the field, smooth the noise and absorb outliers at the same time. This is not a shortcoming of the method, which has been applied successfully to the far richer sample of the Scorpius-Centaurus association [22], but of what the region currently offers. Gaia DR4 will improve exactly this: radial velocities derived from the Radial Velocity Spectrometer are announced for about 312 million sources, and for the first time also at the level of the individual transits [17]. The latter would turn the stellar variability that this work had to absorb into an error term into a measurable quantity. Repeating the reconstruction on that basis is the most direct way to test which of the features described here survive. The model itself also leaves room to grow: a separate scatter term for the radial velocities, which the quality statistics of this work already call for.

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A. Appendix

A.1. Age Histogram

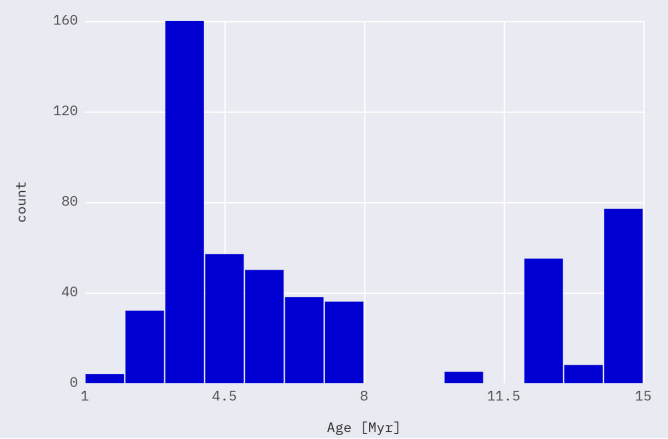


Figure A.1.: Age histogram of 522 stellar sources with available age data.

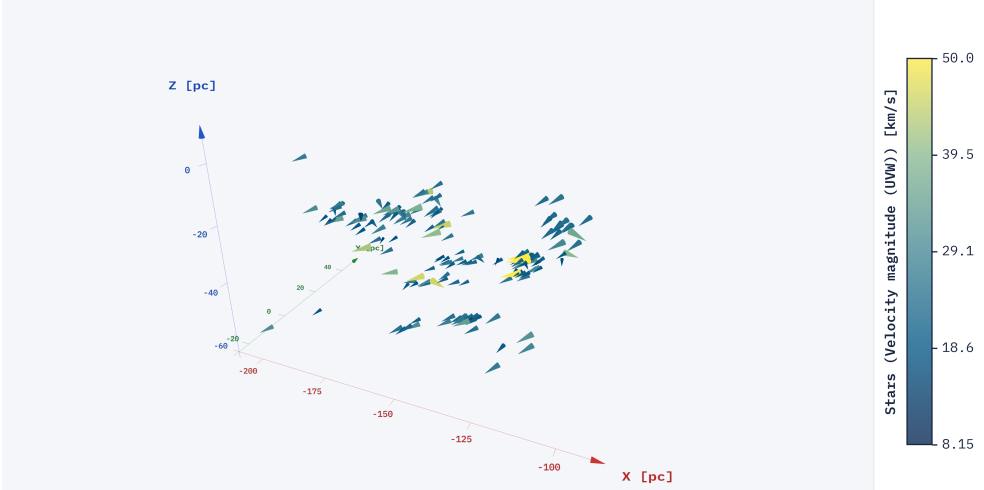
A.2. Velocity Vector Visualizations

Figure A.2a shows the full three-dimensional velocity vectors (U, V, W) of the subsample of 180 stars for which a radial-velocity measurement is available, plotted at their respective three-dimensional positions within the reconstruction volume. Figure A.2b shows the tangential velocities of the full 557-star catalog, drawn as three-dimensional vectors to enable comparison with the full velocity vectors and the velocity field, even though they are only a projection onto the plane of the sky. The sky plane mostly follows the Y–Z plane, as the Sun lies at the origin and Taurus covers only a small patch of sky in the $-X$ direction, so the plotted vectors barely extend in that direction.

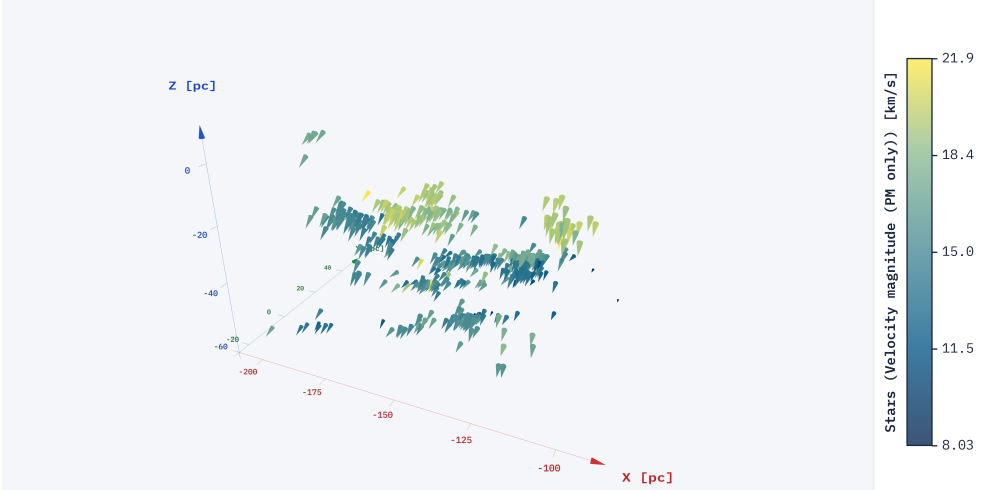
Again, the reader is encouraged to visualise these plots themselves in the interactive viewer to retain a clearer picture than conveyable in these snapshots.

A.3. Density Mask

Figure A.3 shows the stellar mass density underlying the density mask: a Gaussian kernel density estimate of the stellar positions, weighted by the individual stellar masses and



(a) Full three-dimensional velocity vectors of the complete radial-velocity subsample (n=180), shown at their three-dimensional positions. Colorbar clamped at 50 $\mathrm{km\,s^{-1}}$ to account for outliers.



(b) Proper-motion vectors of the full catalog.

Figure A.2.: Vector-based illustrations of the two velocity data modalities entering the reconstruction: full three-dimensional space velocities of the radial-velocity subsample (top) versus proper motions of the full sample, represented as three-dimensional vectors (bottom).

computed with `scipy.stats.gaussian_kde` [36] on the reconstruction grid, scaled by the total mass of the sample to give $\mathrm{M_\odot\,pc^{-3}}$. Masking at any value above zero restricts every figure of the reconstruction to the regions populated by data with variable extent depending on the chosen cut.

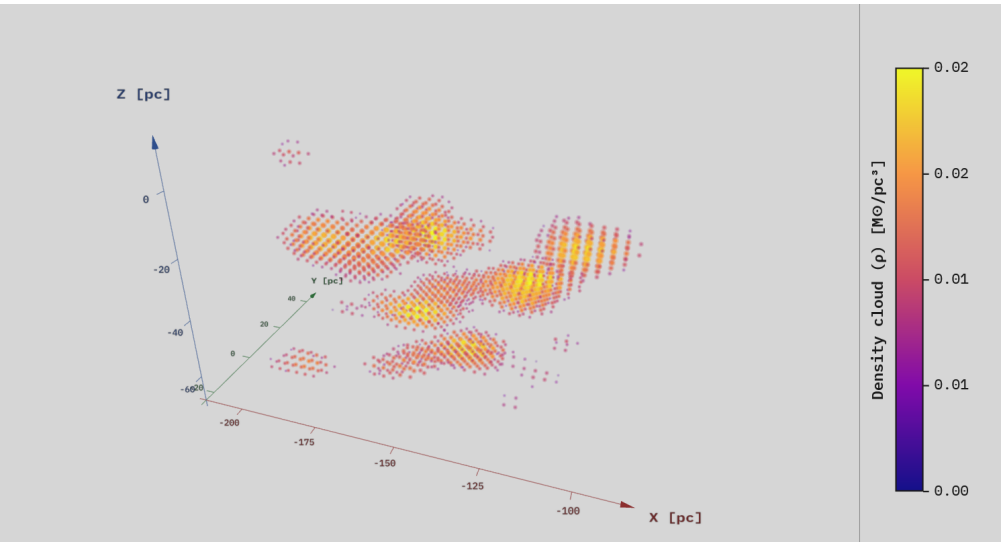
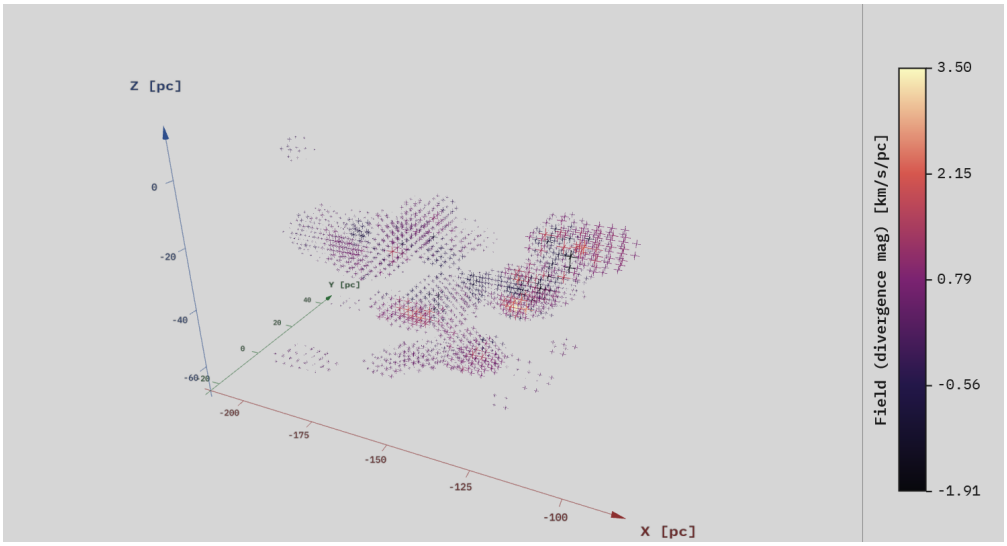


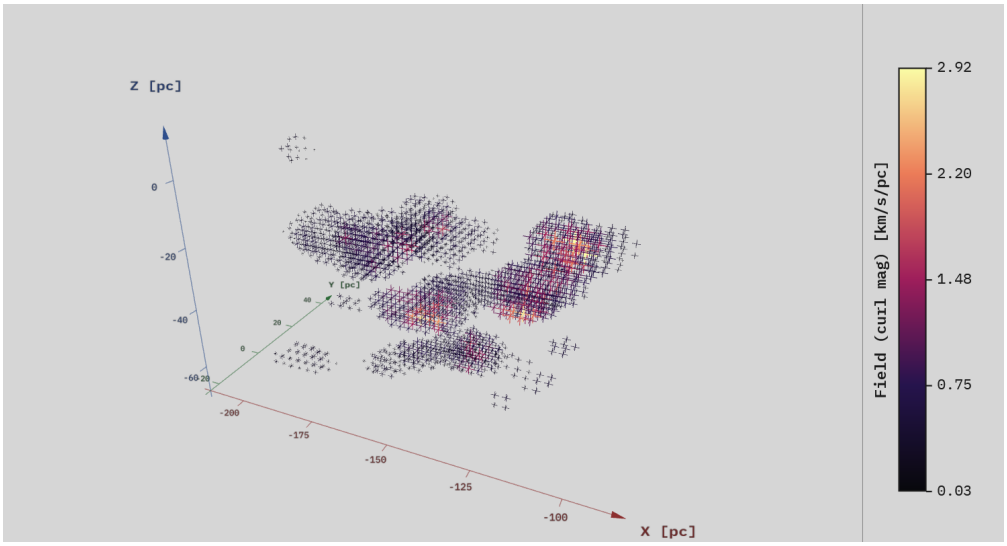
Figure A.3.: Mass density of the stellar dataset. Brighter regions are denser.

A.4. Field Derivatives

Figure A.4 shows the two derived quantities discussed in Section 4.1.2, both in units of $\mathrm{km\,s^{-1}\,pc^{-1}}$ and restricted to the masked region. The divergence is signed: positive values mark regions where the field expands, negative ones where it converges. Of the vorticity only the magnitude $|\nabla \times \vec{v}|$ is shown, as its direction, the axis of the local rotation, is not constrained well enough at this resolution to be meaningfully interpretable.



(a) Divergence $\nabla \cdot \vec{v}$ of the reconstructed field. Positive values indicate expansion, negative values convergence.



(b) Magnitude of the vorticity $|\nabla \times \vec{v}|$ of the reconstructed field.

Figure A.4.: Spatial derivatives of the reconstructed velocity field, computed by central differences on the grid: divergence (top) and vorticity magnitude (bottom).

Software and AI Tools

The field reconstruction was carried out with `NIFTy` [11, 3] and its `JAX` backend [9]. Data handling, coordinate transformations and the kernel density estimates rely on `NumPy` [20], `SciPy` [36], `pandas` [27] and `Astropy` [4]. The interactive vectorfield viewer is built on `three.js` [35].

The AI assistant Claude [2] was used as a tool in three areas: as a programming aid for the analysis code, which started from an existing implementation of the model by Hutschenreuter et al. [22] and was modified and in part reimplemented over the course of this thesis, for the development of the interactive vectorfield viewer, and for language editing of the manuscript, including rephrasing and proofreading. All results, their interpretation and the final wording were checked by the author, who takes responsibility for the content of this thesis.